

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Semester: V

Branch : Electronics and Computer Engineering / Electronics and Computer Science Engineering

Subject Code & Name: Digital Signal & Image Processing (BTECPC502)

Max Marks: 60

Date: 03/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Describe the properties and provide the mathematical expressions for the following elementary signals: impulse, step, and ramp. | CO1 | 6 |
| B) Determine whether the system given by equation linear / non-linear and time variant / invariant i) $y(t) = ax(t) + b$ ii) $y(n) = nx(n)$ | CO1 | 6 |
| C) Classify the signal $x(t) = \cos(2\pi t) + 3\sin(\pi t)$ as even, odd, periodic, non-periodic, deterministic, or non-deterministic, and justify your classification. | CO1 | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Define Region of Convergence (ROC) in the Z-transform? Discuss its properties and explain how the ROC affects the stability and causality of the system. | CO2 | 6 |
| B) What is the Discrete Fourier Transform (DFT)? List and explain at least three properties of the DFT. | CO2 | 6 |
| C) Given two sequences $x[n] = \{1, 2, 3\}$ and $h[n] = \{4, 5\}$ compute their linear convolution using the circular convolution method. Show all steps in your calculation. | CO2 | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Outline the fundamental steps involved in a digital image processing system. Briefly describe each step and its purpose within the system. | CO3 | 6 |
| B) What are the main components of an image processing system? Explain the role of each component and how they interact within the system. | CO3 | 6 |

- C) Discuss some basic image operations, such as image addition, subtraction, and logical operations (AND, OR, NOT) with proper examples. **CO3** **6**

Q.4 Solve Any Two of the following.

- A) Define image enhancement and explain its importance in digital image processing. Provide examples of applications where image enhancement is crucial? **CO4** **6**
- B) What is the Discrete Cosine Transform (DCT) with its significance in image processing, particularly in image compression? **CO4** **6**
- C) Write short note on types of smoothing filters and sharpening filters? **CO4** **6**

Q.5 Solve Any Two of the following.

- A) Write Short notes on Image Segmentation, Edge Detection, thresholding? **CO5** **6**
- B) Given the image A and structuring element B

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Image A

1
1
1

structuring element B

CO5 **6**

Compute A dilated by B

- C) How can hit-or-miss transformation be used for extracting specific pixel configuration in an image? Give suitable example? **CO5** **6**

***** End *****